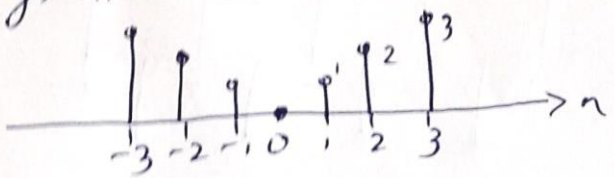
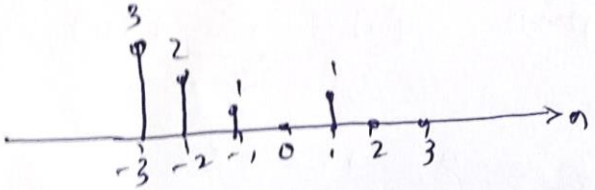
III Sem ECE, Mid term test

Signals & Systems (ECE 2155)

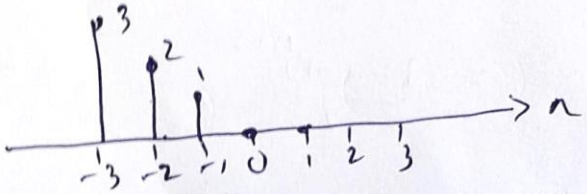
1. $x[n]$ =



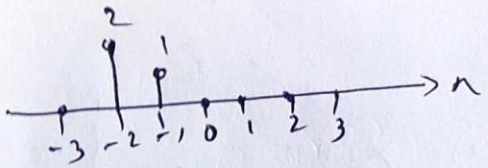

i) $x[n]\,u[1-n]$



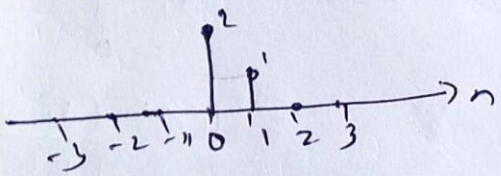

ii) $x[n]\,u[1-2n]$



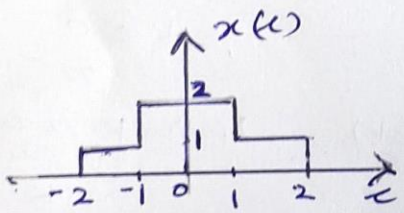

iii) $x[n]\cdot(u[n+2]-u[n])$

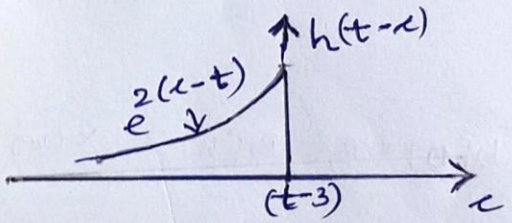



iv) $x[3n-2]\cdot u[-n+3]$




2.

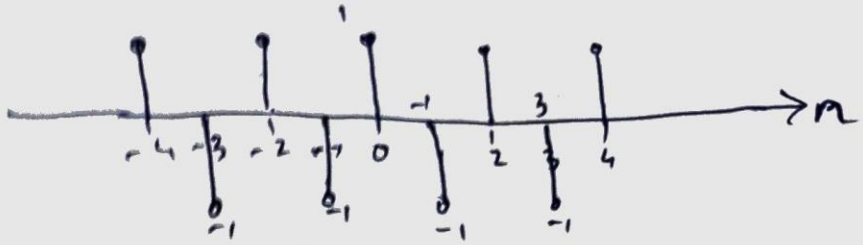



| Range of 't' | $w_t(\tau) = x(\tau)h(t-\tau)$ | output, $y(t)$ |
|---|---|---|
| $t < 1$ | $w_t(\tau) = 0$ | $y(t) = 0$ |
| $1 \le t \le 2$ | $e^{2(\tau-t)}$, $-2 \le \tau \le (t-3)$ | $\frac{1}{2}\left(e^{-6} - e^{-2(t+2)}\right)$ |
| $2 \le t \le 4$ | $e^{2(\tau-t)}$ $-2 \le \tau \le -1$<br>$2e^{2(\tau-t)}$, $-1 \le \tau \le (t-3)$ | $e^{-6} - \frac{1}{2}\left\{e^{-2(t+1)} + e^{-2(t+2)}\right\}$ |
| $4 \le t \le 5$ | $e^{2(\tau-t)}$ $-2 \le \tau \le -1$<br>$2e^{2(\tau-t)}$ $-1 \le \tau \le 1$<br>$e^{2(\tau-t)}$ $1 \le \tau \le (t-3)$ | $\frac{1}{2}\left\{e^{-6} + e^{-2(t-1)} - e^{-2(t+1)} - e^{-2(t+2)}\right\}$ |
| $t > 5$ | $e^{2(\tau-t)}$, $-2 \le \tau \le -1$<br>$2e^{2(\tau-t)}$ $-1 \le \tau \le 1$<br>$e^{2(\tau-t)}$, $1 \le \tau \le 2$ | $\frac{1}{2}\left\{e^{-2(t-2)} + e^{-2(t-1)} - e^{-2(t+1)} - e^{-2(t+2)}\right\}$ |

3.

$h[n]$ :




Step response $S[n] = \sum_{k=-\infty}^{n} h[k]$

$S[n] = 0, \quad n < -4$

$= \frac{1-(-1)^{n+5}}{2}; \quad -4 \le n \le 4$

$= 1, \quad n > 4$

4. i) $X(k) = \frac{1}{T}\int_0^T x(t)\, e^{-jk\cdot\frac{2\pi}{T}t}\, dt \; ; \; \omega_0 = \frac{2\pi}{T}$

$= 4\frac{\sin(k\omega_0 T_0)}{Tk\omega_0} = 2\frac{\sin(k\cdot\omega_0 T_0)}{\pi k}; \quad k \neq 0$

$= 0, \quad k = 0.$

$\therefore B[0] = 0, \quad B[k] = X(k) + X(-k) = 4\frac{\sin\left(k\cdot 2\pi\frac{T_0}{T}\right)}{\pi k}$

$A[k] = 0.$

$x(t) = B[0] + \sum_{k=1}^{\infty} B[k]\cos\left(\frac{2\pi}{T}kt\right) + \sum_{k=1}^{\infty} A[k]\sin\left(\frac{2\pi}{T}kt\right)$

$= \sum_{k=1}^{\infty} B[k]\cdot\cos\left(\frac{2\pi}{T}kt\right)$

ii) System passes cosinesoids of frequency $2\omega_0$ & $3\omega_0$.

( since $\omega_0 = 50\pi$, $H(j\omega) = 1$, $+75\pi \le |\omega| \le 175\pi$

$B[2] = 0, \quad B[3] = -\frac{4}{3\pi}$

$\therefore y(t) = -\frac{4}{3\pi}\cos(150\pi t).$

5. Using Compressed version of $g(t)$ and building the signal (Column wise)

Compress $g(t)$ to get $g(2t)$ and place at different locations, scale and add

∴ $x(t) = g(2t-1) + 2g(2t-3) + 3g(2t-5) + 4g(2t-7)$

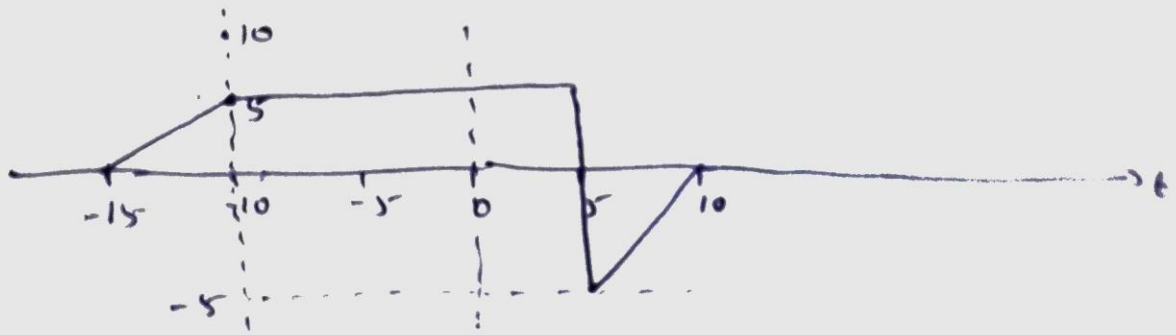
$= g\left(2\left(t-\frac{1}{2}\right)\right) + 2g\left(2\left(t-\frac{3}{2}\right)\right) + 3g\left(2\left(t-\frac{5}{2}\right)\right) + 4g\left(2\left(t-\frac{7}{2}\right)\right)$

Alternatively,

use compressed & expanded $g(t)$ and build (vertically place & step)

$x(t) = g\left(\frac{t}{2}-1\right) + g\left(\frac{3t}{2}-5/3\right) + g(t-3) + g(2t-7)$

6. $x(t) = r(t+15) - r(t+10) - 10u(t-5) + r(t-5) - r(t-10)$




Energy signal, ∴ $E = 2 \times 41.66 + 25 \times 15$

$= 458.33$ units

7. $x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi}\left(\frac{e^{j\Omega} - e^{-j\Omega}}{2j}\right)e^{j\Omega n}\,d\Omega$

$= \frac{1}{2\pi}\cdot\frac{1}{2}\left[\frac{1}{j}\int_{\Omega}\left\{e^{j\Omega(1+n)} - e^{j\Omega(n-1)}\right\}d\Omega\right]$

$= 0, \quad n \neq \pm 1$

$= \frac{1}{2}\cdot\frac{1}{2\pi}\cdot 2\pi = \frac{1}{2}, \quad n = -1$

$= 0 - \frac{1}{2\pi}\cdot\frac{1}{2} = -\frac{1}{2}, \quad n = 1$

$x[n] = \frac{1}{2}\delta(n+1) - \frac{1}{2}\delta(n-1)$

